Question 1 : Correction métier et versioning

Le code modifié et le test unitaire validé et fonctionnel qui vérifie qu'un apprenant ne peut pas s'inscrire à plus de 5 formations.

```php
    /**
     * Règle métier : un apprenant ne peut pas s'inscrire à plus de 5 formations simultanément.
     * Tentative d'une 6ème inscription → 400 Bad Request.
     */
    public function test_apprenant_cannot_enroll_in_more_than_5_formations(): void
    {
        $apprenant = User::factory()->apprenant()->create();

        // Créer 5 formations et inscrire l'apprenant à chacune
        $formations = Formation::factory()->count(5)->create();
        foreach ($formations as $formation) {
            Enrollment::factory()->create([
                'utilisateur_id' => $apprenant->id,
                'formation_id'   => $formation->id,
            ]);
        }

        // Tenter une 6ème inscription
        $sixieme = Formation::factory()->create();

        $response = $this->withSpringAuth($apprenant->email, 'apprenant')
            ->postJson("/api/formations/{$sixieme->id}/inscription");

        $response->assertStatus(400)
            ->assertJsonPath('message', 'Vous ne pouvez pas vous inscrire à plus de 5 formations simultanément.');
    }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

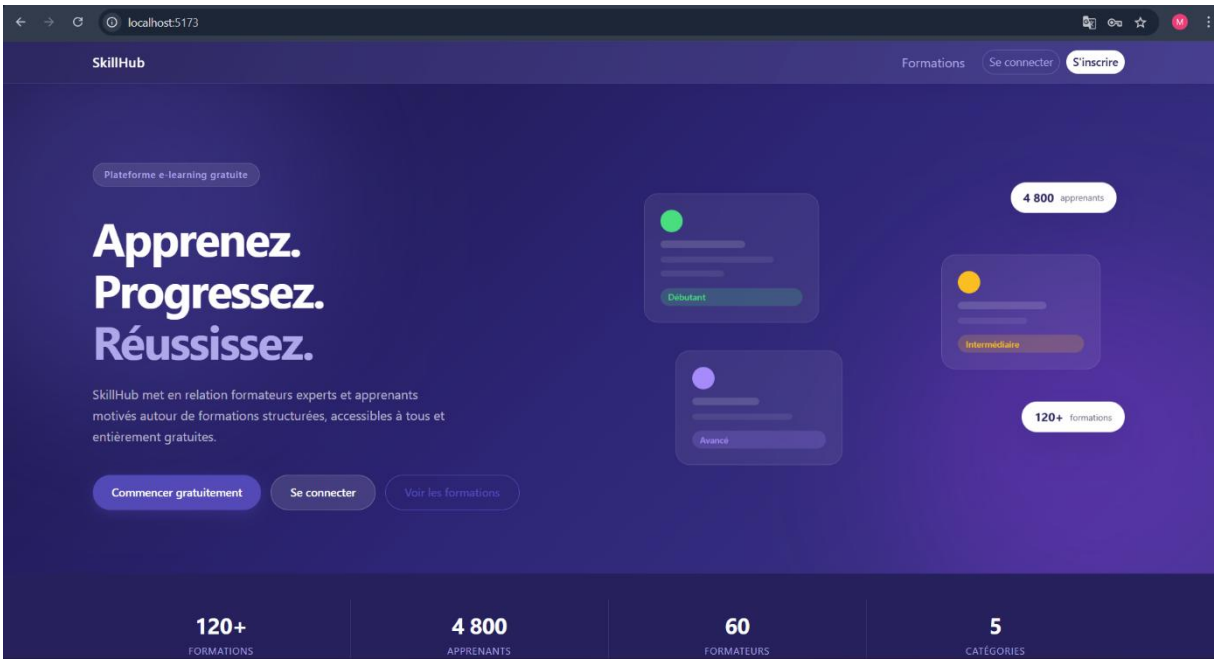
```
C:\skillhub-examen\backend>php artisan test --filter=test_apprenant_cannot_enroll_in_more_than_5_formations

  PASS  Tests\Feature\EnrollmentTest
  ✓ apprenant cannot enroll in more than 5 formations

  Tests:    1 passed (2 assertions)
  Duration: 1.20s
```

Question 2 : Authentification forte avec Spring Boot

Capture de docker compose up qui prouve que tous les services sont démarrés et healthy

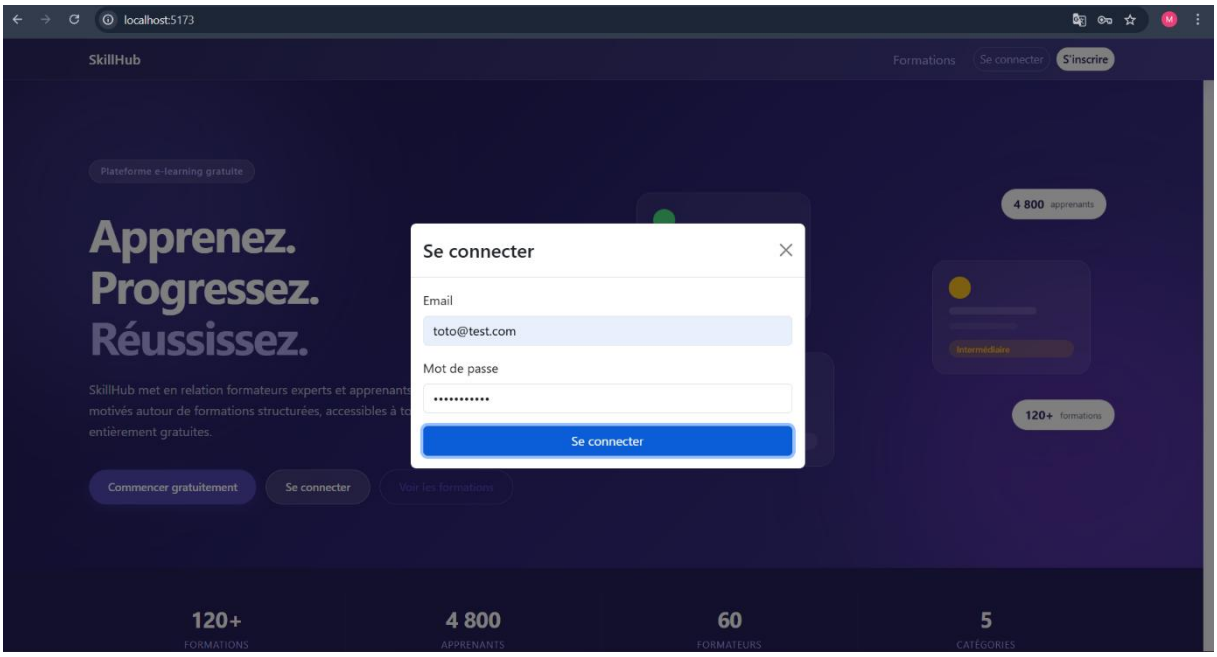
```
C:\skillhub-examen\backend>docker compose up -d
[+] Running 5/5
 ✓ Container skillhub-examen-db-1            Healthy
 ✓ Container skillhub-examen-mongodb-1       Healthy
 ✓ Container skillhub-examen-auth-service-1  Healthy
 ✓ Container skillhub-examen-api-1           Healthy
 ✓ Container skillhub-examen-frontend-1      Started
```

Donc le site est fonctionnel et s’affiche correctement après le docker compose up

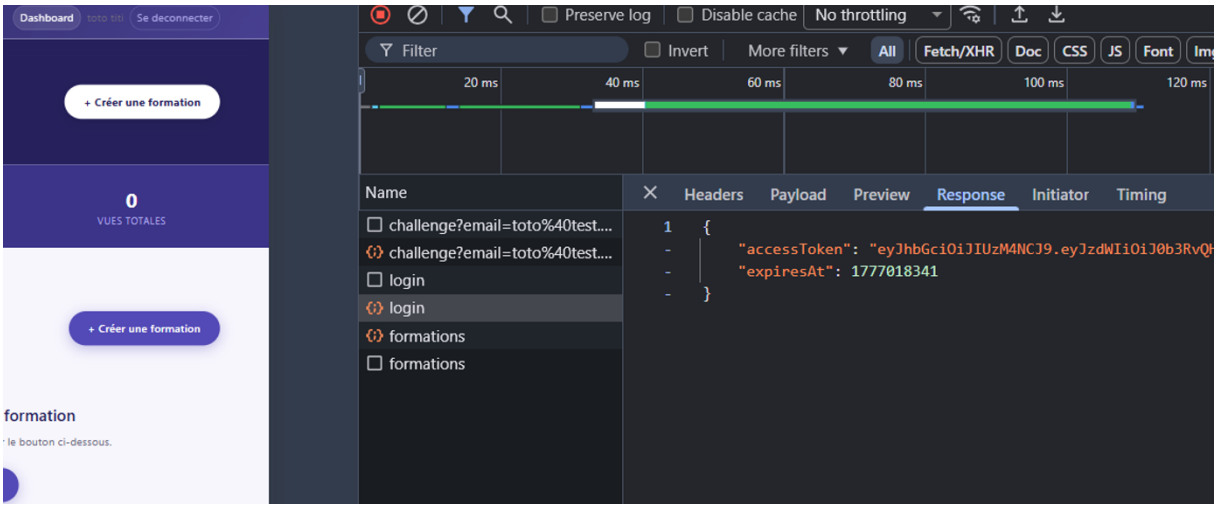


Capture de l’authentification SSO avec Spring Boot :

- Pendant la connexion :

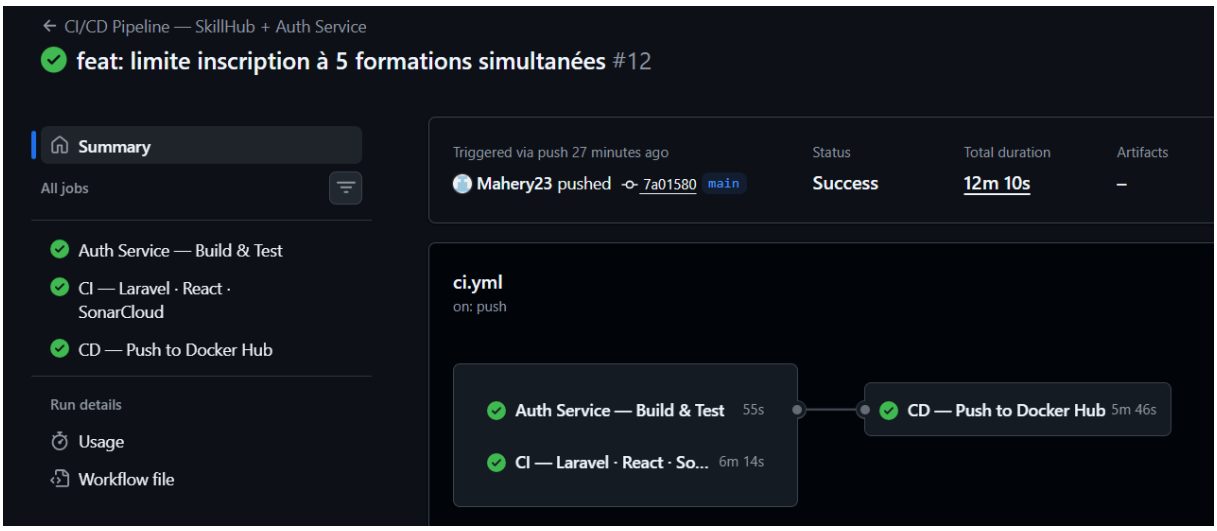


- Après la connexion : on rentre dans le dashboard formateur (route protégée) avec le access token généré par l’authentification SSO de Spring Boot

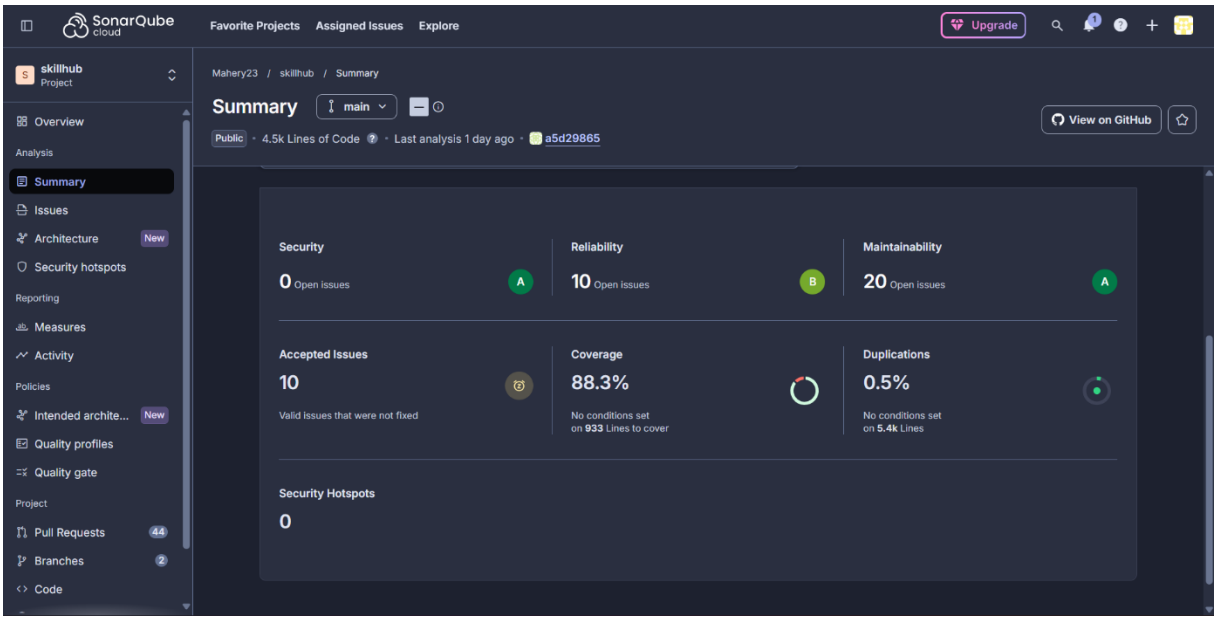


Question 3 : Pipeline CI/CD et Sonarcloud

Pipeline CI/CD fonctionnel qui passe toutes les validations



Capture du rapport de SonarCloud :


SonarQube cloud
Favorite Projects
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Explore
Upgrade
skillhub
Project
Overview
Analysis
Summary
Issues
Architecture
New
Security hotspots
Reporting
Measures
Activity
Policies
Intended archite...
New
Quality profiles
Quality gate
Project
Pull Requests
44
Branches
2
Code
Mahery23 / skillhub / Summary
Summary
main
Public • 4.5k Lines of Code • Last analysis 1 day ago • a5d29865
View on GitHub
Security
0 Open issues
A
Reliability
10 Open issues
B
Maintainability
20 Open issues
A
Accepted Issues
10
Valid issues that were not fixed
Coverage
88.3%
No conditions set on 933 Lines to cover
Duplications
0.5%
No conditions set on 5.4k Lines
Security Hotspots
0